

BossPlz: An Accessible, Multi-Sensory Interactive Virtual Reality (VR) Experience for Extreme Natural Environments

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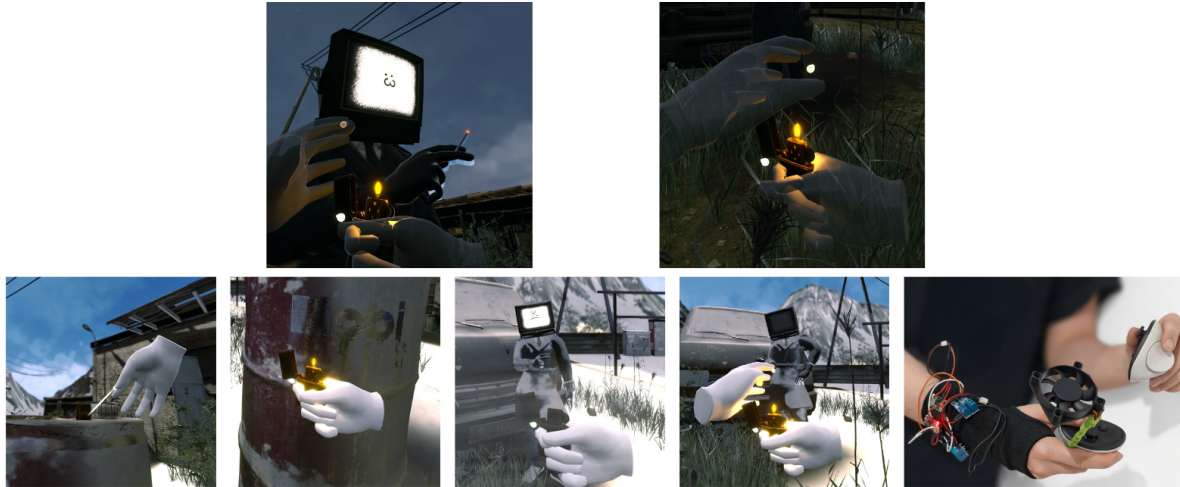


Figure 1: Multi-Sensory VR experience: lighting the lighter for the boss in a natural environment challenge scene.

ABSTRACT

Virtual Reality (VR) simulators often lack physical interference, limiting their use for training in challenging natural or extreme environments. We present a multi-sensory VR system using integrated aero-haptic (wind) and thermal (heat) feedback. Unlike traditional systems focused on passive immersion, our system models environmental forces as an active physical antagonist. The user must perform a fine-motor task (e.g., shielding a flame from wind) while managing this dynamic interference. This work contributes a low-cost, accessible hardware setup and a new interaction paradigm focused on active physical, rather than cognitive, interference management.

Index Terms: Virtual Reality, Thermal Feedback, Bimanual Interaction, Physical Interference, Environmental Simulation.

1 INTRODUCTION

Virtual Reality (VR) offers an accessible pathway to simulate embodied natural and extreme environments, but these simulators often lack physical interference [4]. Physical forces, such as strong winds, directly hinder fine-motor tasks in real environments; however, VR lacks active interference, limiting training application and natural simulations.

Thus, we present BossPlz, a multi-sensory VR experience featuring an accessible, actuator-integrated hardware setup that accommodates temperature and wind interference. In the demo, the user

faces a multi-stage challenge: (1) light a lighter via skill-based precision gesture, (2) transport the lit flame to Boss while physical aero-haptic interference dynamically tries to extinguish it, and (3) use the other hand to shield the flame. This asymmetric bimanual coordination, where each hand serves a distinct role under active physical interference, forms the core of our paradigm (Fig. 1). The contribution of this paper is two-fold:

- A complete, working VR system built on accessible components that integrates precision gesture detection, active aero-haptic interference, and synchronous thermal feedback.
- A novel interaction paradigm that repurposes environmental haptics as an active physical antagonist, requiring embodied asymmetric bimanual coordination under dynamic interference.

Unlike cognitive dual-tasks that degrade performance via attentional bottlenecks, our system introduces *physical* interference as a new paradigm for studying embodied motor coordination. Although demonstrated through a light VR task, this approach directly applies to outdoor training simulations that require fine-motor precision in extreme environmental conditions.

2 RELATED WORK

2.1 From Passive Immersion to Active Challenge

Prior work utilises aero-haptic and thermal feedback primarily to enhance realism, such as simulating speed cues or environmental temperatures [3, 5]. These systems function as tools for *passive immersion*. Although some research explores active challenges through direct force-feedback or somatic penalties (e.g. electrical shocks) [6, 8], BossPlz uniquely repurposes environmental haptics as an *active physical antagonist*. We shift the focus from enhancing presence to creating a physical interference that users must actively manage.

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Figure 2: Haptic-enhanced controller prototype.

2.2 Asymmetric Bimanual Interaction & Interference

Guiard's kinematic chain model [1] establishes a fundamental asymmetry in bimanual action: the non-dominant hand sets a spatial reference frame, while the dominant hand performs fine manipulation within it. Subsequent work confirms that such role differentiation persists during motor learning [2]. Although VR's dual 6-DoF controllers can readily capture this asymmetric coordination, existing VR training paradigms rarely exploit it [7]; instead, they impose *cognitive* interference, secondary memory or attention tasks, that degrade motor performance through attentional bottlenecks rather than direct physical engagement.

3 IMPLEMENTATION

We present a wireless multi-sensory VR system for the Meta Quest 3 with an open-source sensor hardware ecosystem and a custom interaction framework in Unity 3D to deliver active physical interference

3.1 Hardware & System Architecture

The haptic system is powered by an ESP32-C3 microcontroller, chosen for its compact size and direct LiPo battery support (Fig. 2). We integrate a 5V aero-haptic fan and a flexible thermal film, driven by silent MOSFET modules for fast switching. These components are housed in a custom fabric enclosure attached to the Touch Plus controller. Fabric was selected for its light, sustainability, flexibility across hand sizes in research, and ability to diffuse heat safely [9]. The entire add-on uses commodity hardware, costing under \$30 USD and 58g in weight (excluding batteries).

The system follows a host-client model. The Unity host communicates state changes to the ESP32 client via wireless serial connection. A dedicated HapticManager interprets game states (e.g., "Flicker") and transmits binary commands to trigger the corresponding actuators.

3.2 Interaction Logic

The core experience is governed by a Finite State Machine (FSM) (Fig. 3). Haptic feedback is directly coupled to interaction states; for instance, the fan activates only when the lighter is Lit and exposed. A key feature is the "shielding" mechanic, implemented via proximity detection. If the Euclidean distance between the non-dominant hand (shield) and the lighter falls below a 10cm threshold, the flame acts as "covered," inhibiting wind interference. This mechanism translates the user's physical shielding gesture into a system state that mitigates the environmental antagonist.

4 CONCLUSION AND FUTURE WORK

We presented an accessible multi-sensory VR system that repurposes environmental haptics as an active physical antagonist rather than a passive immersion tool, and tested it with a VR scene: BossPlz. Our accessible hardware design (\$30 USD) integrates aero-haptic and thermal feedback for bimanual interference management. Future work will conduct formal user studies under extreme environment simulation to quantify performance differences

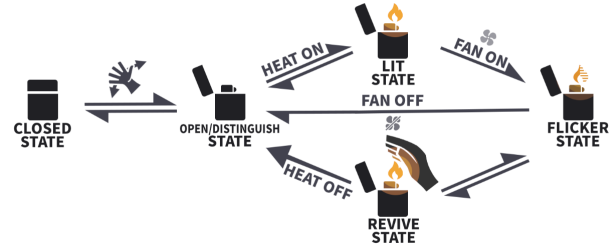


Figure 3: Finite state machine of the virtual lighter.

between physical and cognitive interference paradigms, and explore adaptive difficulty based on real-time physiological sensing.

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